

OIP v1.0.34 — Four-Dataset Audit Synthesis and Evidence Boundary Report

Organic Intelligence Protocol (OIP)

September 2026

Evidence-First | Fail-Closed | Source-Locked | Claim-Disciplined

This document is a descriptive synthesis of four dataset-level audit records. It does not introduce a new dataset, construct operationalization, score, outcome definition, empirical evaluation, or validity claim.

1 Executive Summary

OIP v1.0.34 is a synthesis document rather than a fifth dataset audit. Its scope is the documented lineage from OIP v1.0.26 through v1.0.33, with particular attention to the four dataset-level audits recorded as v1.0.29 (BEMP), v1.0.31 (Malawi IHPS), v1.0.32 (Uganda UNPS 2019/20), and v1.0.33 (Ethiopia ESS4).

Across the supplied audit lineage, the downstream conditions required for authorized construct operationalization and empirical execution were not established in the records available for this synthesis. The documented response was to withhold authorization and preserve the fail-closed boundary rather than fill evidentiary gaps with unsupported proxies, silent imputation, synthetic outcomes, or post-hoc scoring.

The central scientific boundary is narrow: the audited evidence chains define what was not authorized for the datasets examined. They do not establish empirical validity, predictive validity, causal validity, universal validity, superiority, or universal unmeasurability.

2 Purpose and Scope

The purpose of v1.0.34 is to consolidate the methodological and evidentiary record without changing the underlying audit findings.

This document does not select a new dataset, does not operationalize any construct, does not compute any score, and does not claim any empirical validity.

The scope is bounded to the OIP version lineage and audit artifacts identified in this report. Where an artifact is incomplete or unexecuted, that limitation is stated rather than reconstructed from inference.

3 Source Hierarchy and Version Boundary

The governing methodological source is OIP v1.0.30, which defines the evidence hierarchy, construct-qualification requirements, dataset-qualification gates, authorization rule, and claim discipline. OIP v1.0.26 provides the earlier architecture and formal concepts, while v1.0.27 records mathematical provenance and unresolved operationalization questions. OIP v1.0.28 is retained as a historical evaluation record. The later dataset-level records are v1.0.29, v1.0.31, v1.0.32, and v1.0.33.

The v1.0.30 protocol states that the supplied v1.0.26 source contains multiple formula presentations. It records an early summary presentation for the Score expression and a later review-ready specification, and it explicitly prohibits silently choosing between contradictory representations. Therefore v1.0.34 does not retrospectively reconcile source notation. Any confirmatory implementation must follow the version-locked form authorized by the governing protocol.

The evidence hierarchy also excludes AI-generated suggestions from authorization. Candidate suggestions may enter a discovery or review queue, but documentary evidence is required before a variable, construct, transformation, or outcome can be authorized.

4 Documented OIP Architecture

The principal documented stability relation is:

$$\text{OIP Stability} = GFL \times II \times AML = GFL \times (1 - IDS) \times AML.$$

The v1.0.30 mathematical section also records:

$$LIV = \frac{RI \times SR \times AR}{IDS},$$

with the boundary condition

$$IDS = 0 \implies LIV = \text{UNDEFINED}.$$

No denominator replacement, epsilon substitution, clipping, or arbitrary lower bound is authorized for LIV by that protocol.

The v1.0.27 mathematical provenance register additionally records:

$$\text{Score} = 0.6 OE + 0.3 CV + 0.1 VIS,$$

$$SIS = 0.5 RC + 0.3 AFL + 0.2 CAS,$$

and

$$IS = 0.1 SD + 0.9 CW.$$

These are specification-level mathematical structures. Their presence in the documented architecture is not itself evidence of empirical validity or independent external verification.

5 Scientific Authorization Chain

OIP v1.0.30 defines an explicit authorization rule for a future empirical evaluation release. The required conjunction is:

```
Source Definitions Frozen ∧ GFL Approved ∧ IDS Approved ∧ AML Approved ∧
Analytical Cohort Frozen ∧ Score Frozen ∧ Outcome Independent ∧ Outcome
Estimable ∧ Leakage Audit Passed ∧ Evaluation Design Frozen.
```

The same protocol gives the scientific sequence:

$$\text{Evidence} \rightarrow \text{Measurement} \rightarrow \text{Score} \rightarrow \text{Evaluation} \rightarrow \text{Claim}.$$

Its corresponding fail-closed rules are explicit:

- no evidence $\Rightarrow$ no operationalization;
- no approved operationalization $\Rightarrow$ no authorized score;
- no estimable outcome $\Rightarrow$ no authorized binary evaluation;
- no independent evaluation $\Rightarrow$ no empirical validity claim.

For the three later executable audit records with verified final states in the supplied artifacts (Malawi, Uganda, and Ethiopia), the conjunction was not satisfied. The supplied BEMP notebook copy is code-complete in structure but contains no executed cell outputs; its prior runtime state is therefore not reconstructed here from the notebook alone.

6 BEMP Audit — OIP v1.0.29

The v1.0.29 audit targets the Bangladesh Environmental Mobility Panel using the Wave 1 to Wave 14_M evidence chain.

The notebook explicitly preserves missingness as missing, prohibits unsupported proxy mappings, disallows silent imputation and random fallback logic, and requires documentary evidence before construct authorization. It also contains an estimability gate that blocks binary discrimination analysis when only one outcome class is present.

A critical source-integrity limitation remains: the supplied BEMP notebook is an unexecuted notebook artifact. All code cells have no recorded execution output in the supplied copy. Consequently, exact runtime quantities such as a locked-panel size or final positive/negative/unresolved counts cannot be independently verified from this notebook copy. They are intentionally omitted from the synthesis rather than inferred from prior summaries.

Accordingly, this report records the BEMP audit as a source-preserved fail-closed audit lineage, while explicitly declining to manufacture or restate unverified runtime results.

7 Malawi IHPS Audit — OIP v1.0.31

The v1.0.31 audit covers the Malawi Integrated Household Panel Survey for 2010–2019 and documents successful processing of 395 Stata files.

The final technical record reports:

Component	Final state
GFL	PENDING_EVIDENCE
IDS	PENDING_EVIDENCE
AML	PENDING_EVIDENCE
Approved constructs	None
Outcome approval	None
Analytical cohort	NOT_CREATED
Scores	NOT_PERFORMED
Empirical validation	NOT_ESTABLISHED
Final audit status	FAIL_CLOSED_AUDIT_COMPLETE

The audit also distinguishes a documented longitudinal identifier role from actual row-level longitudinal linkage. The record states that unsupported proxy constructs, missing-to-zero conversion, unsupported imputation, synthetic outcome construction, and scoring merely because a formula was computable were not used.

8 Uganda UNPS 2019/20 Audit — OIP v1.0.32

The v1.0.32 notebook defines the dataset as the Uganda National Panel Survey 2019/20, Wave 8, package UGA_2019_UNPS_v03_M_STATA14.

The provenance inventory contains 109 Stata files. The corrected schema audit successfully reads all 109 files. The notebook records zero read errors after the corrected implementation and preserves the distinction between structural integrity and construct authorization.

The final construct states are:

Component	Final state
GFL	NOT_APPROVED
IDS	NOT_APPROVED
AML	NOT_APPROVED
Outcome	NOT_APPROVED
Score authorization	NOT_AUTHORIZED
Empirical evaluation	NOT_READY

The notebook audits 230 candidate outcome variables across several domains. Candidate availability is not treated as outcome authorization. The notebook therefore reaches no authorized score or empirical evaluation.

9 Ethiopia ESS4 Audit — OIP v1.0.33

The v1.0.33 audit covers Ethiopia Socioeconomic Survey 2018–2019, dataset reference `ETH_2018_ESS_v04_M`.

The canonical audit record contains 75 source files: 68 Stata files and 7 PDFs. The metadata layer records 2550 file-variable records and 1625 unique variable names. The value-label layer records 959 file-variable label pairs and 7077 value-label rows. Identifier candidate discovery records 1044 candidate records; the corrected file-variable-level structural audit identifies 23 structurally unique records and 1021 nonunique records.

The documentary evidence stage records 1446 documentary evidence records across the source PDFs. Literal documentary hits are explicitly distinguished from semantic verification and construct authorization.

The final authorization state is:

Component	Final state
GFL	NOT_AUTHORIZED
IDS	NOT_AUTHORIZED
AML	NOT_AUTHORIZED
Outcome	NOT_AUTHORIZED
Score	NOT_AUTHORIZED
Empirical evaluation	NOT_READY
Qualification	FAIL_CLOSED_QUALIFICATION_NOT_ESTABLISHED
Release	RELEASE_AS_FAIL_CLOSED_AUDIT_ONLY

The corrected final manifest reports a SHA-256 value of `efb2e4063baf42a3f78987d6f40063bee39ce7f87108b169f168fbe20d2e3b1f`.

The canonical record also preserves C19 as `BLOCKED`. This is material: the audit did not erase a unit-incommensurability problem merely to obtain a cleaner summary.

10 Cross-Dataset Synthesis Matrix

Version	Dataset	Constructs	Outcome	Score	Empirical
v1.0.29	BEMP	Supplied notebook is code-only; final runtime state not independently reconstructible	Not reconstructed	Not restated	Not restated
v1.0.31	Malawi IHPS	GFL/IDS/AML PENDING_EVIDENCE	Not approved	Not performed	Not established
v1.0.32	Uganda UNPS 2019/20	GFL/IDS/AML NOT_APPROVED	Not approved	Not authorized	Not ready
v1.0.33	Ethiopia ESS4	GFL/IDS/AML NOT_AUTHORIZED	Not authorized	Not authorized	Not ready

The descriptive pattern is that the available evidence did not establish the full conditions required for downstream authorized execution in the audited records. The table deliberately

avoids converting candidate discovery, file readability, or documentary occurrence into construct authorization.

11 Internal Correction and Control Chain — ESS4

The ESS4 notebook contains an explicit sequence of diagnostic, blocking, supersession, and re-run steps. The exact chain is more precise than a simple “correction list”:

Original	Subsequent state	Recorded purpose or consequence
C07	C07R	The structural uniqueness audit was corrected from a variable-level grouping to a file-variable-level unit. C07R then recorded 1044 candidate records, 23 structurally unique and 1021 nonunique.
C09	C09R v1/v2/v3	Intermediate reconciliation attempts were blocked during schema and accounting diagnostics. C09R v3 ultimately passed after exact file-variable-key reconciliation.
C11B	C11D → C11E	Targeted documentary discovery yielded no usable targeted records; safe extraction was blocked when the required engine was unavailable; C11E then used page-wise pypdf extraction and passed with 1446 documentary evidence records.
C18	C18R	C18R preserved the source-locked concept-counting boundary and detected noncanonical terminology. It did not authorize the concepts.
C19	preserved BLOCKED	The final release retained the incommensurability block rather than overwriting it.

The safest interpretation is procedural: the executed ESS4 record contains explicit error detection and state preservation. It does not, by itself, establish a universal claim that the OIP pipeline is self-correcting in every future setting.

12 What the Four Audits Establish

Within the documentary and computational records supplied for this synthesis, the four audit lineage establishes that:

- dataset provenance, file integrity, schema, metadata, documentary occurrence, and evidence states were treated as distinct layers;
- candidate variables were not automatically promoted to authorized constructs;
- file readability was not treated as semantic construct authorization;
- documentary literal hits were not treated as semantic verification;

- downstream score execution was withheld when the required evidence chain was not satisfied in the executable audit records;
- unresolved and blocked states were preserved rather than converted into apparently successful results.

13 What the Audits Do Not Establish

The four audits do not establish:

- empirical validity of OIP;
- predictive validity;
- causal validity;
- universal validity;
- commercial validity;
- superiority over alternative methods;
- that OIP constructs are universally unmeasurable in secondary datasets;
- that primary data collection is universally required.

They also do not justify converting a candidate variable into a construct simply because a numerical result could be obtained from it.

14 Evidence Boundary Statement

The evidence boundary of v1.0.34 is:

Across the dataset-level audits conducted to date within the OIP v1.0.29–v1.0.33 lineage under the OIP v1.0.30 normative protocol, the authorization requirements defined by OIP v1.0.30 were not established for downstream empirical execution in the supplied evidence chains. This defines an evidence boundary for the audited datasets and artifacts. It does not constitute a claim about all secondary data sources, nor a claim that the constructs are universally measurable or universally unmeasurable.

For the current BEMP notebook copy, the exact executed final state is not reconstructed because the supplied notebook contains no executed outputs. This limitation is part of the boundary statement rather than something to be filled by inference.

15 Future Dataset Qualification Criteria (Frozen)

OIP v1.0.30 requires the sequence:

OIP source definitions → measurement criteria → dataset qualification → cohort freeze → evaluation

The protocol explicitly prohibits “dataset first → construct definition later.” Therefore future qualification is to be governed by pre-specified measurement requirements rather than by choosing a dataset because it appears likely to generate a preferred analytical result.

The minimum qualification record should address authoritative definitions, exact variable identity, coding and category meaning, routing and eligibility, missingness provenance, semantic alignment, directionality, transformation requirements, temporal ordering, independent outcome definition, outcome estimability, leakage control, frozen cohort construction, frozen score specification, and evaluation design.

16 Historical v1.0.28 Record

The v1.0.28 report records a preliminary evaluation on an open-source HR benchmark with $N = 54,808$. It reports a baseline promotion yield of 12.66%, a v1.0.28 yield of 20.62%, an absolute difference of 7.96 percentage points, a reported 95% confidence interval of [6.57%, 9.34%], and a reported Welch-test p-value of 3.59×10^{-29} .

These quantities are historical report values and are not presented here as current validation evidence. The report itself describes them as retrospective preliminary empirical evidence and limits their interpretation. In the later OIP lineage, v1.0.28 is classified as a synthetic simulation / decision-logic demonstration and is not treated in this synthesis as independent empirical validation.

No stronger statement such as formal retraction is made here because the supplied source set does not establish that wording as an official status.

17 Reproducibility and Public Artifacts

The research record is organized around versioned source documents, executable notebooks, evidence-state artifacts, and audit manifests. Reproducibility is therefore bounded by the availability of the underlying dataset, source documentation, software environment, and archived outputs.

The relevant public research objects include the v1.0.26 source materials, v1.0.27 mathematical provenance record, v1.0.28 historical report, v1.0.29 BEMP notebook, v1.0.30 normative protocol, and the v1.0.31, v1.0.32, and v1.0.33 audit records.

The ESS4 canonical manifest is identified by the SHA-256 value recorded in Section 9. The BEMP limitation described in Section 6 remains applicable: the current supplied notebook is not an executed result archive.

18 Limitations

The four datasets are not a statistically representative sample of all possible datasets. The synthesis therefore describes the audited evidence chains and does not generalize them to all data sources.

The audits primarily test evidence qualification and authorization. They do not, by themselves, test whether the constructs have predictive, causal, or universal validity.

No new score is introduced by v1.0.34. No new outcome is introduced. No uncertainty estimate is calculated by this synthesis. The historical v1.0.28 statistics are reproduced only as report metadata for lineage transparency.

The ESS4 audit contains preserved blocked and not-run states. In particular, the statement `NO_SYNTHETIC_OUTCOME_CONSTRUCTION = NOT_ESTABLISHED` must not be misread as evidence that a synthetic outcome was constructed; it records that the qualification gate itself was not formally established.

19 Current Scientific Position and Claim Boundary

The current evidence-bounded position is:

The OIP research lineage contains four dataset-level evidence audits. The supplied executable audit records for Malawi, Uganda, and Ethiopia document withheld construct and/or downstream authorization under the OIP v1.0.30 rules. The supplied BEMP notebook preserves the relevant fail-closed code but does not contain executed outputs, so its exact runtime final state is not reconstructed here. Accordingly, v1.0.34 is a descriptive evidence-boundary record, not a proof of OIP effectiveness, ineffectiveness, validity, or universal applicability.

The methodological rule remains unchanged:

Evidence must authorize measurement; measurement must authorize score; score must precede evaluation; evaluation must precede claim.